

A Short Guide to Lemma D.1

Why a covariant quantum operation has constant success probability

Abstract

Lemma D.1 says that a displacement-covariant, trace-nonincreasing quantum operation can lose only a fixed fraction of the trace: its success probability cannot depend on the input state. The proof is a short operator-theoretic version of Schur's lemma. This note explains the duality behind the proof, the irreducibility of the Weyl representation, and the lemma's role in the Gaussian stochastic-order argument.

1 The content in one sentence

Let Γ be a possibly probabilistic operation on one or finitely many oscillator modes. If displacing the input by z merely displaces the output by $\sqrt{\gamma}z$, then no displacement can change the probability that the operation succeeds. Because the Weyl displacement representation is irreducible, this invariance is strong enough to make the success probability the same for *every* input state, not only for states on one displacement orbit.

Equivalently, there is a single number $c \in [0, 1]$ such that

$$\mathrm{Tr} \Gamma(\rho) = c$$

for every normalized state ρ . When $c > 0$, the rescaled map $c^{-1}\Gamma$ is a deterministic channel.

2 Background

2.1 Trace class and bounded operators

For $s < \infty$ oscillator modes, let

$$\mathcal{H} = \mathcal{F}^{\otimes s}, \quad \mathcal{T}_1(\mathcal{H}) = \{\text{trace-class operators on } \mathcal{H}\}, \quad \mathcal{B}(\mathcal{H}) = \{\text{bounded operators on } \mathcal{H}\}.$$

The two spaces are in duality:

$$\mathcal{T}_1(\mathcal{H})^* = \mathcal{B}(\mathcal{H}), \quad \langle X, A \rangle = \mathrm{Tr}(XA).$$

Thus every bounded linear functional ℓ on the trace class has a unique representing operator $A \in \mathcal{B}(\mathcal{H})$:

$$\ell(X) = \mathrm{Tr}(AX). \tag{1}$$

Moreover, ℓ is positive exactly when $A \geq 0$.

If $\Gamma : \mathcal{T}_1(\mathcal{H}) \rightarrow \mathcal{T}_1(\mathcal{H})$ is bounded, its Banach adjoint is the map

$$\Gamma^* : \mathcal{B}(\mathcal{H}) \longrightarrow \mathcal{B}(\mathcal{H}), \quad \mathrm{Tr}[\Gamma(X)A] = \mathrm{Tr}[X\Gamma^*(A)].$$

In particular, the functional recording the output trace is represented by the *effect*

$$A = \Gamma^*(I), \quad \text{Tr } \Gamma(X) = \text{Tr}(AX). \quad (2)$$

If Γ is completely positive and trace nonincreasing, then

$$0 \leq A \leq I.$$

Indeed, positivity gives $A \geq 0$, while $\text{Tr}[\Gamma(X)] \leq \text{Tr } X$ for every $X \geq 0$ gives $A \leq I$ by duality. For a normalized input ρ , $\text{Tr}(A\rho)$ is precisely the success probability of the operation.

2.2 Weyl displacements

On one mode,

$$D(z) = \exp(za^* - \bar{z}a), \quad z \in \mathbb{C}.$$

On s modes, $z = (z_1, \dots, z_s) \in \mathbb{C}^s$ and

$$D(z) = \bigotimes_{j=1}^s D(z_j).$$

The Weyl relation contains a scalar phase,

$$D(z)D(w) = e^{i\text{Im}(z \cdot \bar{w})} D(z + w),$$

but this phase disappears under conjugation. Consequently, $X \mapsto D(z)XD(z)^*$ is an ordinary action of the additive phase-space group.

The Weyl representation on the s -mode Fock space is irreducible. Its commutant is therefore

$$\{A \in \mathcal{B}(\mathcal{H}) : AD(z) = D(z)A \text{ for every } z \in \mathbb{C}^s\} = \mathbb{C}I. \quad (3)$$

This is the Stone–von Neumann irreducibility statement, followed by Schur’s lemma. Informally, commuting with all position and momentum translations leaves an operator no nontrivial degree of freedom.

2.3 Displacement covariance

The covariance condition in the paper is

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^*. \quad (4)$$

The factor $\sqrt{\gamma}$ is the amplitude gain. Its value will not enter the proof: only the fact that the output transformation is unitary, and hence trace preserving, is needed.

3 Statement of Lemma D.1

Lemma D.1 (rephrased). *Let $\Gamma : \mathcal{T}_1(\mathcal{H}) \rightarrow \mathcal{T}_1(\mathcal{H})$ be completely positive and trace nonincreasing, and suppose that (4) holds for every $z \in \mathbb{C}^s$. Then there is a constant $c \in [0, 1]$ such that*

$$\text{Tr } \Gamma(X) = c \text{Tr } X \quad (5)$$

for every trace-class operator X .

The statement includes subnormalized and nonpositive X . It is enough to prove the identity first for positive X ; linearity then extends it to all trace-class operators.

4 Proof, line by line

Step 1: encode the output trace by one operator

Let $A = \Gamma^*(I)$. By (2),

$$\mathrm{Tr} \Gamma(X) = \mathrm{Tr}(AX),$$

and trace nonincrease gives $0 \leq A \leq I$. Thus all possible input-dependence of the success probability is stored in the single effect A .

Step 2: covariance makes the effect invariant

Take the trace of both sides of (4). Unitary invariance of the trace gives

$$\mathrm{Tr} \Gamma(D(z)XD(z)^*) = \mathrm{Tr} \Gamma(X).$$

Using the representation by A and cyclicity of the trace,

$$\begin{aligned} \mathrm{Tr}[AD(z)XD(z)^*] &= \mathrm{Tr}(AX), \\ \mathrm{Tr}[(D(z)^*AD(z) - A)X] &= 0. \end{aligned}$$

This holds for every trace-class X . Since trace class separates bounded operators,

$$D(z)^*AD(z) = A \quad (z \in \mathbb{C}^s). \quad (6)$$

Step 3: irreducibility forces a scalar

Equation (6) says that A belongs to the Weyl commutant. By (3), $A = cI$ for some scalar c . The order bounds $0 \leq A \leq I$ imply $0 \leq c \leq 1$. Substitution into (2) yields

$$\mathrm{Tr} \Gamma(X) = \mathrm{Tr}(cIX) = c \mathrm{Tr} X,$$

which is (5).

This also covers the endpoint cases. If $c = 1$, Γ is trace preserving. If $c = 0$, then $\Gamma(X)$ is a positive trace-class operator of trace zero for every $X \geq 0$, hence $\Gamma = 0$.

5 Intuition and examples

The effect A is the quantum analogue of a state-dependent acceptance function. Covariance says this function cannot distinguish any two points connected by a phase-space displacement. The representation is irreducible, so there are no hidden invariant sectors on which different acceptance probabilities could be assigned. Therefore A must be constant.

For example, if $\mathcal{A}_\gamma$ is the quantum-limited amplifier, then

$$\Gamma = c\mathcal{A}_\gamma, \quad 0 \leq c \leq 1,$$

satisfies the lemma. It performs the amplifier with an input-independent probability c . By contrast, a filter that accepts according to the vacuum effect,

$$\mathrm{Tr} \Gamma(\rho) = \langle 0|\rho|0\rangle,$$

is not displacement covariant: displacing the state changes its vacuum overlap.

Irreducibility is essential. If a symmetry representation decomposed into two invariant sectors, an effect of the form $c_1P_1 + c_2P_2$ would commute with the symmetry while assigning different success probabilities to the two sectors. Weyl displacements exclude this possibility.

6 Dependencies and role in the paper

The proof uses only three standard facts:

1. trace-class/bounded-operator duality, which gives $A = \Gamma^*(I)$;
2. invariance and cyclicity of the trace;
3. irreducibility of the finite-mode Weyl representation.

It does not use the Gaussian fidelity bound or the stochastic-order lemmas that follow it.

Lemma D.1 is used immediately in Lemma D.3. There one starts with a trace-nonincreasing covariant map. Lemma D.1 writes it as

$$\Gamma = c \Gamma_0,$$

where either $c = 0$ and the claim is trivial, or $\Gamma_0 = c^{-1}\Gamma$ is a trace-preserving displacement-covariant amplifier. The Guță–Matsumoto least-noise theorem applies to Γ_0 . Multiplication by c then turns their normalized stochastic-order statement into the conic inequality needed for a subnormalized output. Lemma D.1 is therefore the bridge between probabilistic limit maps and the usual theory of deterministic Gaussian amplifiers.

7 Minimal prerequisite checklist

To read Lemma D.1, it is enough to know that a completely positive trace-nonincreasing map represents a possibly probabilistic quantum operation, that its success effect is $\Gamma^*(I)$, and that the multimode Weyl displacement representation is irreducible. The proof is precisely Schur’s lemma applied to the success effect.